GitHub reproducible workbook

YU-LIN CHUNG, 35480351

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INTRODUCTION

This is an reproducible workbook to demonstrated the guide of knowledge of Git and GitHub usage.

WORKING GUIDELINE

1. Creating a new RStudio Project and Quarto file

First, you need to create a new RStudio Project called "ETC5513_Assignment2" and use the following command to create a simple qmd file called example.qmd.

- The command `cd` is used to move your head into the project folder.

```
cd ETC5513_Assignment2
```

- The command `touch filename.qmd` create a new empty Quarto file.

```
touch example.qmd
```

- To make it be knitted into a HTML file, you need to write the YAML format of html and render it.

```
---
title: "GitHub reproducible workbook"
author: "YU-LIN CHUNG, 35480351"
format:
    html:
        toc: true
        theme: solar
---
```

- The command `ls` will list all of the file in "ETC5513_Assignment2". It can be seen that we successfully knitted the file into HTML when there is a file call "example .html"

```
zhongyulin@TKs-MacBook ETC5513_Assignment2 % ls
ETC5513_Assignment2.Rproj       example.html                    guide_files                     guide.qmd
example_files                   example.qmd                     guide.html
```

2. Initialising the folder as a Git repository

After creating and rendering example.qmd, we can use the following command to initialise the project folder as a Git repository.

- The command `git init` create a new local Git repository in the project folder.

```
git init
```

- The command `git add` will add the change into the staging area. The dot means the current directory. THerefore, `git add .` stages all changes in the current project folder.

```
git add .
```

- The command `git commit` create the first commit in the repository. We usually add the message with `-m` to remind us for our implement.

```
git commit -m "Inital update"
```

3. Create the new branch and add the changes

Now we need to do two things: First, create a new branch called testbranch. Second, modify the file example.qmd and add the changes to both the local and remote repositories.

- To make sure we are in the main branch. The command `git branch` can ensure where you are if you see an asterisk symbol in fornt of the branch name.

```
git branch
```

- The command `git switch` will create a new branch and switch you to this branch.

```
git switch -c testbranch
```

- After modifying example.qmd, we can use `git status` to check the status of it.

```
git status
```

- Following the above steps, we need to add the modified file to the staging area with the command `git add` and commit the change to the local repository with the command `git commit` and write the message for it.

```
git add example.qmd
git commit -m "Modify example.qmd on testbranch"
```

- Finally, we need to push the new branch and its commit to the remote GitHub repository by using command `git push`. However, because it is the first time to push the new branch to GitHub, we need to set up upstream with `-u`

```
git push -u origin testbranch
```

4. Add the data and amend the previous commit

Next, we need to create a folder called data, and add the data from Assignment 1 to that folder.

- We need to made sure we are working on testbranch. The command `git branch` will show where you are with the asterisk symbol in fornt of the branch name.

```
git branch
```

- The command `mkdir` create a new folder.

```
mkdir data
```

- The command `cp` is used to copy the data in to the folder "data" .

```
cp "/Users/zhongyulin/assignment-1-creating-reproducible-reports-yulinchung/data/35480351_Yu
```

- The command `cd` is used to move into sepcific folder and we can use `ls` to check that the file had been copied successfully.

```
cd data
ls
```

After that, we will learn how to amend the previous commit to include the data folder and push this amended commit to the remote.

- Instead of creating a new commit, we need to amend the previous commit. The command `git commit --amend` modifies the most recent commit. The option `-m` allows us to write a new commit message.

```
git commit --amend -m "Modify example.qmd and add data folder"
```

- Finally, we need to push it to the remote GitHub repository by using command `git push`. However, because we amended a commit that had already been pushed, the commit history changed. Therefore, we need to use `git push -f` rather than a normal `git push` to force GitHub to update the remote branch.

```
git push -f origin testbranch
```

5. Create merge conflict

- Typing some similar content in main branch as the same position in testbranch.

```
### Changes of the testbranch
Change of the new branch call "testbranch"
```

Figure 1: content in testbranch

```
### Changes of the main branch
Change made on the main branch to create a conflict with testbranch.
```

Figure 2: content in main

6. Deal with merge conflict

- After completing the steps above, we will find that the system reminds us there is a conflict.

```
zhongyulin@TKs-MacBook ETC5513_Assignment2 % git merge testbranch
warning: Cannot merge binary files: .DS_Store (HEAD vs. testbranch)
Auto-merging .DS_Store
CONFLICT (content): Merge conflict in .DS_Store
Auto-merging example.qmd
CONFLICT (content): Merge conflict in example.qmd
Automatic merge failed; fix conflicts and then commit the result.
zhongyulin@TKs-MacBook ETC5513_Assignment2 %
```

- We need to fix the merge conflict.

```
## INTRODUCTION
This is an example .qmd for my assignment2.

Accept Current Change | Accept Incoming Change | Accept Both Changes | Compare Changes
<<<<<<< HEAD (Current Change)
### Changes of the main branch
Change made on the main branch to create a conflict with testbranch.
=======
### Changes of the testbranch
Change of the new branch call "testbranch"
>>>>>>> testbranch (Incoming Change)
```

```
## INTRODUCTION
This is an example .qmd for my assignment2.

### Changes from both branches
This section combines the changes made on branches of main and testbranch. The
main branch was modified to create a merge conflict and testbranch was modified
earlier to demonstrate branch-based changes.
```

- After resolving the conflict, we can use `git status` to check the current status. This command shows whether all conflicts have been fixed and whether the files are ready to commit.

```
git status
```

- After the conflict was fixed, we used `git add` to add the changes into the staging area. Then, we commit the resolved merge conflict. This commit records the final version after merging testbranch into main. Finally, we push the merge commit to the remote GitHub repository.

```
git add example.qmd
git commit -m "Resolve merge conflict"
git push
```

- After pushing, we can refresh the GitHub repository page and check the `main` branch. The latest commit message is `Resolve merge conflict`, which shows that the merge result has been pushed to GitHub.

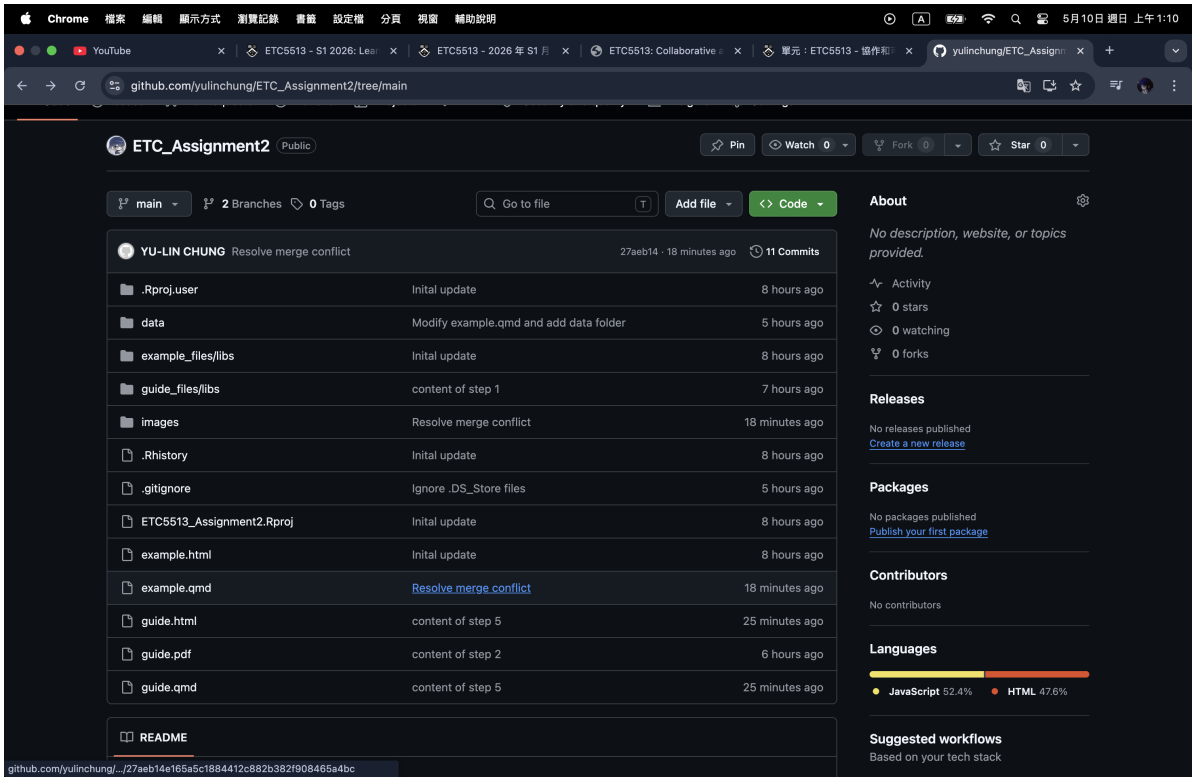

Figure 3: the GitHub status